Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/18/2012
PI Name:	Kozlowski, Steve Ph.D.		
Project Title:	Developing, Maintaining, and Restoring Team Cohesion		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	I moved from Michigan State University to the University of South Florida in August 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	08/16/2009	**End Date:**	08/15/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:	Leveton, Lauren	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/15/2013 per NSSC information (Ed., 7/13/2012)		
Key Personnel Changes/Previous PI:	(June 2012 report) Neal Schmitt, Leaetta Hough, and James Locke were originally tasked to conduct research with analog teams in the Antarctic. However, efforts to gain access via NSF have been stalled. These research team members are currently inactive. Dr. Daisy Chang has joined the project as Co-I.		
COI Name (Institution):	DeShon, Rick (Michigan State University) Biswas, Subir (Michigan State University) Schmitt, Neal (Michigan State University) Chang, Chu-Hsiang (Daisy) (Michigan State University)		
Grant/Contract No.:	NNX09AK47G		
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Task Description:

Team cohesion, collaboration, and psycho-social health are essential ingredients for team effectiveness, particularly for those “high reliability” action teams that perform critical tasks under the stress of isolated, confined, and extreme (ICE) environments – such as teams of astronauts. Although there is considerable research that has established a relationship between team cohesion and team effectiveness, there is very little research that examines how teams cohere and collaborate over time.
This ground-based research is designed to accomplish three goals:

(1) develop interventions to enhance the development and maintenance of team cohesion; (2) create psychological and physiological measurement systems that monitor team psycho-social health and trigger dynamic interventions to maintain it and restore it; and (3) provide recommendations for (a) critical knowledge and skills (i.e., teamwork competencies, relationship management) and (b) monitoring/regulating interventions for psycho-social health maintenance.

Project goals will be accomplished using a multidisciplinary team and a combination of research strategies:

(1) Laboratory research using a team simulation will be used to (a) examine interventions to enhance team functioning and (b) develop measurement techniques for assessing team collaboration and cohesion. (2) Prototype sensor-based monitoring systems to assess team collaboration and psycho-social functioning. (3) Promising interventions and monitoring/regulation techniques will be examined in the field with long duration teams operating in extreme environments to evaluate proof of concept.

The research is designed to resolve IRP Gap BHP 2.3.1 by identifying training content, design, and delivery to build and maintain team cohesion and BHP 2.2.1 developing a monitoring system designed to support, regulate, and repair team cohesion and maintain performance.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Team cohesion is not just a critical factor for astronaut teams and ground crews; cohesion is important to the effectiveness of all teams and especially those that operate in critical, high reliability setting. Of the many team process factors that support team effectiveness, team cohesion is the most studied with over a half century of research. Yet, remarkably, very little is known about the characteristics that promote its development and maintenance. For example, we know that experience working together is associated with cohesion formation and maintenance, but what are the mechanisms? Teams that do not cohere replace problematic members or disintegrate so experience only reveals those teams that survive, but that does not tell us why or how. This research, which will uncover the dynamics of collaboration, cohesion, and effective team functioning and will create technologies to monitor team cohesion and guide interventions to restore it, has the potential for wide utility in aviation, military, medical, industrial, and other environments where society depends on the effective performance of high reliability teams.

Project Organization
The project is organized around the 3 primary research foci:

• The Simulation Team (Optima Lab) is tasked with developing team simulations where we can examine collaborative teamwork with precision and prototyping behavioral metrics to capture “micro-cohesion” and other indicators that capture the behavioral dynamics of team member interaction during complex task performance. Our initial research focused on team collaboration during a computer-based exploration simulation Mars Acquisition of Resources Simulation (MARSim). That research established the value and viability of behavioral metrics as important indicators and contributors to team performance. Subsequent research has focused on validating metrics from the Monitoring Technology (below) and developing an integration of multimodel, dynamic team collaboration data streams.

o Key personnel include Dr. Steve Kozlowski, PI; Dr. Rick DeShon, Co-I; Dr. Daisy Chang; and research assistants Samantha Baard, Mike Braun, Marina Pearce, and Tara Rench.

• The Monitoring Technology Team is tasked with developing hardware and software to enable wireless monitoring of team member collaboration and engagement during task performance. This year, work has focused on hardware and software development of (a) a “diarization” system for social conversation monitoring, (b) a swallow monitoring system (to monitor food and drink intake, arousal), (c) adding a Bluetooth interface to enhance data collection flexibility, and (d) a cell phone App to serve as a mobile “dashboard” for monitoring signals.

o Key personnel include Dr. Subir Biswas, Co-I; and research assistants Yogesh Piolet and Dong Bo.

• The Analog Team is tasked with developing access to analog teams (i.e., long duration, isolated, extreme conditions) to conduct benchmarking research to examine relationships between individual differences; team collaboration, cohesion, an d conflict; and team effectiveness.

o Key personnel for the original effort included Neal Schmitt, Leaetta Hough, and James Locke. However, they encountered problems gaining access to the analog environment. Their effort is on indefinite hold.

o Given those difficulties, an alternative team led by the PI assumed responsibility for this research thrust in year 2 (2011). Key personnel include Dr. Steve Kozlowski, PI; Dr. Rick DeShon, Co-I; Dr. Daisy Chang; and research assistants Samantha Baard, Mike Braun, Marina Pearce, and Tara Rench.

Research Progress

Simulation Research Team

Laboratory Validation of Sensor Data and Wireless Network – Face Time Collaboration

A team simulation for collaborative interaction was developed based on adaptations to the NASA Space Flight Resource Management (SFRM) task “Moon Base.” To provide an appropriate validation platform, the lab simulation was designed to provide a task context that necessitated highly structured interactions. Since our initial validation efforts are designed to assess the reliability and consistency of the interaction data streamed by the wireless monitoring technology, structured interactions facilitate data analysis and inference.

A full laboratory experiment was conducted with 123 undergraduate students forming 41 three person teams. Team members were equipped with monitoring technology “badges,” which stream interaction data to a computer server. Teams participated in a series of interactions throughout a one hour period in order to analyze the algorithm of face time collaboration identified through the initial validation study (year 2). Video recordings of these interactions were coded by three raters blind to the conditions to provide an independent data stream of objective interactions to validate these algorithms and the monitoring technology.

1,742 interactions were analyzed, comparing both the sensor data and independent video coding. Results indicated that the sensor data correctly identified the individuals in the interaction with 99% accuracy, who initiated an interaction with 99% accuracy, and who disengaged from an interaction with 97% accuracy. Furthermore, the sensors correctly identified the total length of the interactions within several seconds of the video coding with 98% accuracy.

These validation results suggest that the sensors and algorithms are well calibrated and accurate in identifying face time collaborations. Next research steps will begin to reduce the degree of structure imposed on the interactions as we continue our phased validation process for the monitoring technology.

Laboratory Validation of Sensor Data and Wireless Network – Affect (in progress)

A second laboratory experiment is in progress to examine and validate the sensor data with regard to affective responses. This research added a “stressor” to the study design (time pressure) and relaxed interaction constraints somewhat. We also collected dynamic assessments of affective reactions for each interaction from the participants. Data collection is complete. Algorithms are currently being developed to identify patterns of affect throughout the interactions. Video recordings and experience sampling methodology are being used to provide a validation for the algorithms.

Monitoring Technology Team

Development of Audio Processing Algorithms for Speaker Detection Using Wearable Sensors

In this task, we have developed an algorithm for speaker diarization using wearable wireless sensors. The key challenges of developing such algorithms is to be able to perform experimental acoustic diarization for social conversation monitoring in the presence of low acoustic data sampling rates, noise and variable inter-personal distances. The key idea is to collect acoustic sensor data from strategically placed wireless sensors over conversing human subjects’ bodies, and to collate and process that data using an Acoustic Comparator based Diarization (ACD) algorithm in order to identify the speaker in a conversation at any given time. Through experiments, with body mounted sensors, it is demonstrated that ACD algorithms can be used for effective speaker identification with high precision. Controlled experiments using human subjects were carried out for evaluating the performance of the developed ACD algorithms. Swallow Monitoring Through Apnea Detection in Breathing Signal

Task Progress:

In this task we have developed the concepts, design, and algorithms for a wearable swallow monitoring system. Swallow monitoring can be used for assessing a person’s psychological state, overall food and drink intake habits, and other swallow related disorders. The system works based on the key observation that a person’s otherwise continuous breathing process is interrupted by a short apnea when she or he swallows as a part of the food intake process. Using a wearable chest-belt, we detect swallows by detecting apneas extracted from breathing signals captured by the chest-belt. A filter-based template matching framework is developed for swallow detection. A number of matched filter template designs, both static and iterative, are developed for high-accuracy swallow detection.

Adding Bluetooth Interface for Collecting Data from the Wearable Sensor to Data-Enabled Cell Phones

The first version of the system designed in this project uses a 900MHz access point for collecting data from a wearable sensor and then feeding into a notebook/desktop computer for processing and dissemination over wide area network. While this model worked for the initial set of experiments, it suffers from the limitation that a subject, wearing a sensor, always needs to be within the range of the access point (several meters), thus, limiting the utility of the system in wide areas and outdoor settings.

The main goal for developing a Bluetooth interface to the wearable sensor in this task is to be able to collect sensor data from a regular data-enabled mobile phone/device, thus eliminating the dependency on access points. This will extend the utility of the system in outdoor settings. A Bluetooth interface kit EZ430-RF256x from Texas Instrument is used for connecting the sensor to mobile phones. The Bluetooth link is now active and live.

Developing a Cell Phone App for Rendering and Managing Wearable Sensor Information

We started developing an Android App for the mobile phone which acts as a mobile dashboard to render and manage the information from the wearable sensor collected through a Bluetooth link as described in Task 3. This phone App will reproduce all functions that are available in the desktop GUI version of the data rendering system that is currently being used. Data collected by the App can be uploaded to a server in both off-line and in real-time using WiFi and 3G connections. This development expands the utility of the monitoring system.

Analog Research Team

Benchmark collaboration, cohesion, and conflict dynamics in long duration analog teams

During 2011, we collected data from analog teams operating in an isolated, confined, extreme environment (the Antarctic) with characteristics similar to those experienced by astronauts in space. Although the sample size was somewhat small, we were able to examine team members working collectively over the course of 42 days – on every day between the start of a mission to its end. This allowed us to learn about how collaborative interactions (i.e., cohesion and conflict) fluctuates over time in two different teams and what other factors influence (and are influenced by) those dynamics.

We were fortunate to have the opportunity to collect another round of data from this source in 2012. This past winter, another team of scientists participated in a 48-day mission on the ice in Antarctica. We gathered similar data as we did last year in an effort to extend our benchmarking results. That is, using pre-expedition and post-expedition surveys, as well as daily diaries for team members and leaders, we assessed a number of key variables including: Individual differences (i.e., demographics and educational experience), daily thoughts and reactions regarding taskwork and teamwork, and member satisfaction and performance. As in the previous data collection effort, we used qualitative and quantitative measures in order to gain the most detail possible.

	Consistent with our findings in 2011, this new team's cohesion also varied substantially over time. While cohesion ratings tended to be favorable overall, the pattern became more stable toward the latter half of the season. In addition, members tended to disagree about how cohesive the team was nearer the beginning of the mission; toward the middle of the mission, they started consistently converging that the team was cohesive. In order to examine the underlying causes of this pattern of cohesion, we read through team members' open-ended diary responses to identify its potential influencers. One important influencer of cohesion, particularly during the first half of the mission, was the boredom and frustration experienced as a result of being confined to tents for long stretches of time (due to weather). This prevented team members from completing their mission (collecting meteorites) and from working together toward common goals. Many times, it seemed this resulted in sub-teams working on separate tasks or individual team members engaging in activities on their own. This may be one reason why members provided relatively weaker ratings of cohesion – and disagreed about how cohesive their team was – during the first half of the mission. Research Summary A substantial amount of cross-sectional (i.e., static) research has shown a strong relationship between team cohesion and team performance. Although team cohesion is generally assumed to be a relatively stable team process variable, little is known about its potential dynamics. Across different types of benchmarking data (military teams during 8 weeks of intensive training), simulation teams (across hours to weeks of performance), problem solving teams (hours), and analog teams (6 weeks deployed in an ICE environment), we observed considerable variation in team cohesion and team collaboration effectiveness. This convergent evidence across very different teams, tasks, and timeframes makes it evident that the assumed stable relationship between team cohesion and team effectiveness needs to be examined closely and unpacked. This research is developing metrics, tools, and techniques to do just that. Schedule Status Key tasks are on schedule and the project is making good progress. There have been some delays, however, that should be noted. Initial project start-up was delayed by administrative difficulties. Essentially, it took 2 months to get an account set up; this delay in the provision of funds for the project slowed somewhat our initial progress. In addition, there two areas that merit discussion. First, we have refocused our simulation research to concentrate on behavioral assessments of cohesion and collaboration. This is important, in our view, for developing a suite of assessments to complement the monitoring technology. Second, access to analog environments for both (a) baseline assessments of team functioning over long duration in ICE environments and (b) evaluation of monitoring technology / metrics continues to be a challenge.
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